

UFICAP - Ciência da Computação - 2º período

Disciplina: Elementos da Integração Computacional

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### Lista #13 - Revisão p/ prova

① Calcule:  $\int_0^3 (2x^2 - 5x + 4) dx = 2 \int_0^3 x^2 dx - 5 \int_0^3 x dx + 4 \int_0^3 dx$

$$= \frac{2}{3} x^3 \Big|_0^3 - \frac{5}{2} x^2 \Big|_0^3 + 4x \Big|_0^3 = \frac{2}{3} \cdot 27 - \frac{5}{2} \cdot 9 + 4 \cdot 3 = 18 - \frac{45}{2} + 12 = 36 - 45 + 24 = 21$$

② Avalie:  $\int_1^2 \frac{\ln x}{x} dx$

$\frac{dv}{dx} = \frac{1}{x} \quad \frac{dx}{dv} = x$

$$= \int_0^1 v dv = \frac{v^2}{2} \Big|_0^1 = \frac{1}{2}$$

③ Calcule:  $\int_0^\pi \sin x dx = -\cos x \Big|_0^\pi = -\cos \pi + \cos 0 = -(-1) + 1 = 2$

④ Resolva:  $\int_0^1 x e^{-2x} dx$

$\begin{cases} v = x \\ \frac{dv}{dx} = 1 \end{cases} \quad \begin{cases} u = e^{-2x} \\ \frac{du}{dx} = -2e^{-2x} \end{cases}$

$\int v \frac{du}{dx} dx = \frac{1}{-2} \int \frac{dv}{du} du = -\frac{1}{2} \int \frac{1}{u} du = -\frac{1}{2} \ln |u| + C$

$= x \cdot \frac{-1}{2} e^{-2x} - \int \frac{-1}{2} e^{-2x} dx = -\frac{1}{2} x e^{-2x} + \frac{1}{2} \int e^{-2x} dx$

$$= -\frac{1}{2} x e^{-2x} + \frac{1}{2} \left( -\frac{1}{2} e^{-2x} \right) \Big|_0^1 = -\frac{1}{2} x e^{-2x} - \frac{1}{4} e^{-2x} \Big|_0^1$$

$$= -\frac{1}{2} \cdot 1 \cdot e^{-2} - \frac{1}{4} \cdot e^{-2} - \left( -\frac{1}{2} \cdot 0 \cdot e^0 - \frac{1}{4} \cdot e^0 \right) = -\frac{1}{2} e^{-2} - \frac{1}{4} e^{-2} + \frac{1}{4}$$

$$= -\frac{1}{2} \left( e^{-2} + \frac{1}{2} e^{-2} - \frac{1}{2} \right)$$

⑤ Calcule:  $\int_1^4 \frac{1}{x} dx = \ln x \Big|_1^4 = \ln 4 - \ln 1 = \ln 4 = \ln 4$



6) Encontre o comprimento da curva:  $y = 2x^{3/2} - x$ ;  $0 \leq x \leq 1$

$$\frac{dy}{dx} = 3x^{1/2} - 1 = \sqrt{3x} - 1$$

$$\int_a^b \sqrt{1 + [f'(x)]^2} dx$$

$$\therefore \int_0^1 \sqrt{1 + (3\sqrt{x} - 1)^2} dx = \int_0^1 \sqrt{1 + 9x - 6\sqrt{x} + 1} dx$$

$$= \int_0^1 \sqrt{9x - 6\sqrt{x} + 2} dx$$

$u = \sqrt{x} \Rightarrow x = u^2 \Rightarrow \frac{dx}{du} = 2u \Rightarrow dx = 2u du$

$$= \int_0^1 \sqrt{9u^2 - 6u + 2} \cdot 2u du$$

$$= 2 \int_0^1 \sqrt{9u^2 - 6u + 2} du = \int_0^1 \sqrt{9u^2 - 6u + 2} du$$

$$= 9 \left( u^2 - \frac{2}{3}u + \frac{1}{9} \right) + 2 = 9 \left( \left( u - \frac{1}{3} \right)^2 - \frac{1}{9} \right) + 2$$

completa o quadrado:  $\frac{1}{9} - \frac{2}{3} = -\frac{1}{3}$

3 1ºs termos são um quadrado perfeito!

$$\left( \frac{-1}{3} \right)^2 = \frac{1}{9}$$

3) soma e subtrai por 1/9

$$= 9 \left( u - \frac{1}{3} \right)^2 - 1 + 2 = 9 \left( u - \frac{1}{3} \right)^2 + 1$$

objetivo  $a^2 + b^2 \rightarrow t^2 + 1$

$$\therefore \int_0^1 \sqrt{9 \left( u - \frac{1}{3} \right)^2 + 1} du = \int_0^1 \sqrt{t^2 + 1} dt$$

$\frac{t+1}{3} = \frac{1}{3}(t+1) = \frac{1}{3}t + \frac{1}{3}$

$$\therefore \int_0^1 \sqrt{t^2 + 1} dt = \int_{-1}^1 \sqrt{t^2 + 1} dt$$

$$= 2 \int_0^1 \sqrt{t^2 + 1} dt = 2 \cdot \frac{1}{3} \int_0^1 (t+1) \sqrt{t^2 + 1} dt = \frac{2}{9} \int_{-1}^1 (t+1) \sqrt{t^2 + 1} dt$$

$$= \frac{2}{9} \left[ \int_{-1}^1 (t \sqrt{t^2 + 1} + \sqrt{t^2 + 1}) dt \right] = \frac{2}{9} \left[ \int_{-1}^1 t \sqrt{t^2 + 1} dt + \int_{-1}^1 \sqrt{t^2 + 1} dt \right]$$

$$= \frac{2}{9} \left[ \frac{1}{3} (5\sqrt{5} - 2\sqrt{2}) + \int_{-1}^1 \sec^2 y dy \right]$$

1)  $w = t^2 + 1$   
 $\frac{dw}{dt} = 2t \Rightarrow dw = 2t dt$   
 $\frac{dw}{2} = t dt$   
 $\frac{1}{2} \cdot \frac{1}{3/2} \cdot w^{3/2} \Big|_{-1}^1$   
 $= \frac{1}{3} \cdot \frac{1}{3/2} \cdot \frac{1}{2} \cdot \frac{3/2}{3/2} \Big|_{-1}^1$   
 $= \frac{1}{3} \cdot \frac{1}{3/2} \cdot \frac{1}{2} \cdot \frac{3/2}{3/2} \Big|_{-1}^1$

$$= \frac{1}{2} (\sec y \tan y + \ln |\sec y + \tan y|) + C$$

2)  $t = \tan y$   
 $\tan^2 y + 1 = \sec^2 y$   
 $dt = \sec^2 y dy$

$$\therefore \int \sec^2 y \sec y dy = \int \sec^3 y dy = \frac{1}{3} (5\sqrt{5} - 2\sqrt{2})$$



$$\therefore \frac{2}{9} \left[ \frac{1}{3} (5\sqrt{5} - 2\sqrt{2}) + \frac{1}{2} (\sec y \tan y + \ln |\sec y + \tan y|) \right] \Big|_0^2 \rightarrow \text{voltage } 1 \oplus \dots$$

$$= \frac{2}{9} \left[ \frac{1}{3} (5\sqrt{5} - 2\sqrt{2}) + \frac{1}{2} (\sqrt{t^2+1} \cdot t + \ln |\sqrt{t^2+1} + t|) \right] \Big|_0^2 \quad \begin{matrix} t = \tan y \\ \sqrt{t^2+1} = \sec y \end{matrix}$$

$$= \frac{1}{2} (\sqrt{2^2+1} \cdot 2 + \ln |\sqrt{2^2+1} + 2|) - \frac{1}{2} (\sqrt{(-1)^2+1} \cdot (-1) + \ln |\sqrt{(-1)^2+1} + (-1)|)$$

$$= \frac{1}{2} (2\sqrt{5} + \ln(2+\sqrt{5})) - \frac{1}{2} (-\sqrt{2} + \ln(\sqrt{2}-1)) = \frac{1}{2} [2\sqrt{5} + \ln(2+\sqrt{5}) - (-\sqrt{2} + \ln(\sqrt{2}-1))]$$

$$= \frac{1}{2} (2\sqrt{5} + \ln(2+\sqrt{5}) + \sqrt{2} - \ln(\sqrt{2}-1))$$

$$\therefore \frac{2}{9} \left[ \frac{1}{3} (5\sqrt{5} - 2\sqrt{2}) + \frac{1}{2} (2\sqrt{5} + \sqrt{2} + \ln(2+\sqrt{5}) - \ln(\sqrt{2}-1)) \right] \text{ u.c.}$$

⑦ A curva é dada por:  $x(t) = t$ ;  $y(t) = t^2$ ;  $0 \leq t \leq 2$

Calcule o comprimento do arco.

\* Arco param:

$$\therefore \int_0^2 \sqrt{1^2 + (2t)^2} dt = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$\begin{matrix} t = \tan y \\ dt = \frac{1}{2} \sec^2 y dy \end{matrix} \quad \left( \frac{1}{2} \int_0^2 \sqrt{\sec^2 y} \cdot \sec^2 y dy = \frac{1}{2} \int_0^2 \sec^3 y dy \right) \rightarrow \frac{1}{2} (\tan y \sec y + \ln |\tan y + \sec y|)$$

$$= \frac{1}{2} \cdot \frac{1}{2} (\tan y \sec y + \ln(\tan y + \sec y)) \Big|_0^2 \rightarrow \left| \tan y = 2t \right. \\ \left. \sec y = \sqrt{1 + \tan^2 y} = \sqrt{1 + 4t^2} \right.$$

$$= \frac{1}{4} (2t\sqrt{1+4t^2} + \ln(2t + \sqrt{1+4t^2})) \Big|_0^2$$

$$= \frac{1}{4} (2 \cdot 2\sqrt{1+4 \cdot 2^2} + \ln(2 \cdot 2 + \sqrt{1+4 \cdot 2^2})) - \frac{1}{4} (2 \cdot 0\sqrt{1+4 \cdot 0^2} + \ln(2 \cdot 0 + \sqrt{1+4 \cdot 0^2}))$$

$$= \frac{1}{4} (4\sqrt{17} + \ln(4 + \sqrt{17})) - \frac{1}{4} (\ln(1)) \rightarrow \frac{1}{4} (4\sqrt{17} + \ln(4 + \sqrt{17})) \text{ u.c.}$$

$$(\ln \sqrt{1} = \ln 1 = 0)$$

⑧ Determine o comprimento da curva:  $y = \cosh x$ ;  $0 \leq x \leq 1$ .

$$\frac{dy}{dx} = \sinh x \therefore \int_0^1 \sqrt{1 + (\sinh x)^2} dx = \int_0^1 \sqrt{\cosh^2 x} dx = \int_0^1 \cosh x dx \quad \text{p/ } \cosh x \geq 0$$

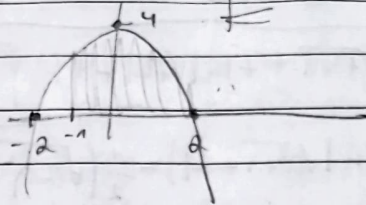
$$= \sinh x \Big|_0^1 = \sinh 1 - \sinh 0 = \sinh 1 \quad \text{ou } \frac{e^1 - e^{-1}}{2} \text{ u.c.}$$



⑨ Encontre a área entre:  $y = 4 - x^2$  e o eixo  $X$ , no intervalo  $[-1, 2]$ .

$y=0$

$$4 - x^2 \leq 0 \rightarrow x^2 \geq 4 \rightarrow x \geq 2 \text{ ou } x \leq -2$$



$$\int_{-1}^2 (4 - x^2 - 0) dx = \int_{-1}^2 4 dx - \int_{-1}^2 x^2 dx$$

$$= 4x \Big|_{-1}^2 - \frac{x^3}{3} \Big|_{-1}^2 = 4 \cdot 2 - \frac{2^3}{3} - \left( 4 \cdot (-1) - \frac{(-1)^3}{3} \right) = 8 - \frac{8}{3} + 4 - \frac{-1}{3} = \frac{24 - 8 + 12 - 1}{3}$$

$$= \frac{27}{3} = 9 \text{ u.a.}$$

⑩ Calcule a área entre as curvas:  $y = e^x$  e  $y = x + 2$ .

4 Pontos de interseção  $\approx -1,8414$  e  $1,1461$

$$\int_{-1,84}^{1,14} (x + 2 - e^x) dx = \int_{-1,84}^{1,14} x dx + \int_{-1,84}^{1,14} 2 dx - \int_{-1,84}^{1,14} e^x dx = \frac{x^2}{2} + 2x - e^x \Big|_{-1,84}^{1,14}$$

$$= \frac{(1,14)^2}{2} + 2 \cdot 1,14 - e^{1,14} - \left( \frac{(-1,84)^2}{2} + 2 \cdot (-1,84) - e^{-1,84} \right)$$

$$\approx \frac{1,2996}{2} + 2,28 - 3,1267 - \left( \frac{3,3856}{2} - 3,68 - 0,1588 \right)$$

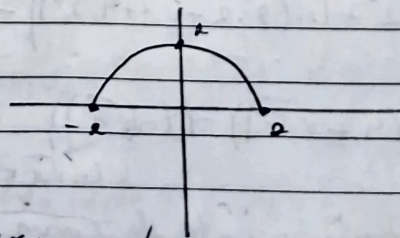
$$\approx 0,6498 + 2,28 - 3,1267 - 1,6928 + 3,68 + 0,1588 \approx 1,9499 \text{ u.a.}$$

⑪ Determine a área da região:  $y = \sqrt{4 - x^2}$ ;  $-2 \leq x \leq 2$ .

$$\int_{-2}^2 \sqrt{4 - x^2} dx$$

$$\sqrt{a^2 - x^2}$$

→ metade superior de um círculo;



4 Fórmula:

$$\int \sqrt{a^2 - x^2} dx = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \arcsin\left(\frac{x}{a}\right) + C$$

$$\therefore \int_{-2}^2 \sqrt{4 - x^2} dx = \frac{x}{2} \sqrt{4 - x^2} + 2 \arcsin\left(\frac{x}{2}\right) \Big|_{-2}^2$$

$$= \left( \frac{2}{2} \sqrt{4 - 2^2} + 2 \arcsin\left(\frac{2}{2}\right) \right) - \left( \frac{(-2)}{2} \sqrt{4 - (-2)^2} + 2 \arcsin\left(\frac{-2}{2}\right) \right)$$

$$= 2 \arcsin 1 - 2 \arcsin(-1) = 2\pi \text{ u.a.}$$



12) Volume gerado por:  $y = x^3$ ;  $0 \leq x \leq 1$ , girado em torno do eixo  $X$

$$4\pi \int_0^1 (x^3)^2 dx = \pi \int_0^1 x^6 dx = \pi \frac{x^7}{7} \Big|_0^1$$

$$= \pi \cdot \frac{1}{7} - 0 = \frac{\pi}{7} \text{ u.v.}$$

$$V = \pi \int_a^b y^2 dx$$

13) Volume por cascas cilíndricas:  $y = 2x - x^2$ ;  $x \in [0, 2]$ , girado em torno do eixo  $y$ .

$$4V = 2\pi \int_0^2 x(2x - x^2) dx = 2\pi \int_0^2 (2x^2 - x^3) dx$$

$$V = 2\pi \int_a^b x f(x) dx$$

$$= 2\pi \left( 2 \int_0^2 x^2 dx - \int_0^2 x^3 dx \right) = 2\pi \left( \frac{2x^3}{3} - \frac{x^4}{4} \Big|_0^2 \right)$$

$$= 2\pi \left( \frac{2}{3} \cdot 2^3 - \frac{2^4}{4} \right) = 2\pi \left( \frac{16}{3} - 4 \right) = 2\pi \cdot \frac{16-12}{3} = 2\pi \cdot \frac{4}{3} = \frac{8\pi}{3} \text{ u.v.}$$

14) Volume gerado pela rotação da região sob:  $y = \sqrt{x}$ ;  $0 \leq x \leq 4$ , em torno do eixo  $X$ .

$$4\pi \int_0^4 (\sqrt{x})^2 dx = \pi \cdot \frac{x^2}{2} \Big|_0^4 = \pi \cdot \frac{4^2}{2} = \frac{8\pi}{1} \text{ u.v.}$$

15) A variável aleatória  $X$  tem densidade:

$$f(x) = \frac{1}{5}; \quad 0 \leq x \leq 5. \text{ Calcule:}$$

a)  $P(1 \leq X \leq 3)$ :  $\int_1^3 \frac{1}{5} dx = \frac{1}{5} x \Big|_1^3 = \frac{3}{5} - \frac{1}{5} = \frac{2}{5}$

b)  $E(X)$ :  $\int_0^5 x \cdot \frac{1}{5} dx = \frac{1}{5} \cdot \frac{x^2}{2} \Big|_0^5 = \frac{1}{10} \cdot \frac{x^2}{2} \Big|_0^5 = \frac{25}{10} = \frac{5}{2}$

16) A variável aleatória possui densidade:  $f(x) = \frac{x}{8}$ ;  $0 \leq x \leq 4$ .

a)  $P(x < 2) \rightarrow [0, 2]$

$$\int_0^2 \frac{x}{8} dx = \frac{1}{8} \frac{x^2}{2} \Big|_0^2 = \frac{1}{16} x^2 \Big|_0^2 = \frac{4}{16} = \frac{1}{4}$$

b)  $E(X) \rightarrow [0, 4]$

$$\int_0^4 x \cdot \frac{x}{8} dx = \frac{1}{8} \int_0^4 x^2 dx = \frac{1}{8} \frac{x^3}{3} \Big|_0^4 = \frac{1}{24} x^3 \Big|_0^4 = \frac{4^3}{24} = \frac{64}{24} = \frac{8}{3}$$



17) O tempo de operação tem densidade:  $f(t) = kt^2 e^{-t/2}$ ,  $t \geq 0$ .

a) Determine  $k$ :  $\int_0^{\infty} kt^2 e^{-t/2} dt = 1$ ;  $k \int_0^{\infty} t^2 e^{-t/2} dt = 1$

$\therefore k \left( t^2 \cdot -2e^{-t/2} - \int_0^{\infty} -2e^{-t/2} \cdot 2t dt \right) = 1$

$k \left( -2t^2 e^{-t/2} + 4 \int_0^{\infty} t e^{-t/2} dt \right) = 1$

$\therefore k \left( -2t^2 e^{-t/2} + 4 \left( -2te^{-t/2} + 2 \int_0^{\infty} e^{-t/2} dt \right) \right) = 1$

$k \left( -2t^2 e^{-t/2} + 4 \left( -2te^{-t/2} - 4e^{-t/2} \right) \right) \Big|_0^{\infty} = 1$

$k \left( -2t^2 e^{-t/2} - 8te^{-t/2} - 16e^{-t/2} \right) \Big|_0^{\infty} = 1 + k \left[ -2e^{-t/2} (t^2 + 4t + 8) \right] \Big|_0^{\infty} = 1$

$k \left[ \lim_{a \rightarrow \infty} \left( -2e^{-a/2} (a^2 + 4a + 8) \right) - \left( -2e^{-0/2} (0 + 0 + 8) \right) \right] = 1$

$k \cdot 16 = 1$

$k = \frac{1}{16}$

b)  $P(T > 3)$ :  $\int_3^{\infty} \frac{1}{16} \cdot t^2 e^{-t/2} dt = \frac{1}{16} \int_3^{\infty} t^2 e^{-t/2} dt$

$= \frac{1}{16} \left[ -2e^{-t/2} (t^2 + 4t + 8) \right] \Big|_3^{\infty} = \frac{1}{16} \left[ \lim_{a \rightarrow \infty} \left( -2e^{-a/2} (a^2 + 4a + 8) \right) - \left( -2e^{-3/2} (3^2 + 4 \cdot 3 + 8) \right) \right]$

$= \frac{1}{16} \left[ 2e^{-3/2} (9 + 12 + 8) \right] = \frac{1}{16} \cdot 58e^{-3/2} = \frac{29}{8} e^{-3/2}$

18) Densidade normalizada:  $f(h) = \frac{h}{18}$ ;  $0 \leq h \leq 6$ .

a)  $P(2 \leq H \leq 5)$ :  $\int_2^5 \frac{h}{18} dh = \frac{1}{18} \cdot \frac{h^2}{2} \Big|_2^5 = \frac{1}{36} h^2 \Big|_2^5 = \frac{1}{36} \cdot 5^2 - \frac{1}{36} \cdot 2^2$

$= \frac{25}{36} - \frac{4}{36} = \frac{21}{36} = \frac{7}{12}$

b)  $\text{Var}(H)$ :  $E(H^2) - (E(H))^2 \rightarrow E(H) = \int_0^6 \frac{h}{18} \cdot h dh = \frac{1}{18} \int_0^6 h^2 dh = \frac{1}{18} \cdot \frac{h^3}{3} \Big|_0^6$

$= \frac{1}{54} h^3 \Big|_0^6 = \frac{1}{54} \cdot 6^3 = \frac{216}{54} = 4$



$$E(H^2) = \int_0^6 h^2 \cdot \frac{h}{18} dh = \frac{1}{18} \int_0^6 h^3 dh = \frac{1}{18} \frac{h^4}{4} \Big|_0^6 = \frac{1}{72} h^4 \Big|_0^6$$

$$= \frac{1}{72} 6^4 = \frac{1296}{72} = 18 \quad \therefore \text{Var}(H) = 18 - 4^2 = 12 //$$

19) A taxa de crescimento  $\hat{r}$ :  $r(t) = 200 e^{-0,1t}$  //

Calcule o crescimento total em 12 meses.

$$\int_0^{12} 200 e^{-0,1t} dt = 200 \int_0^{12} e^{(0,1t)} dt$$

$u = 0,1t$   
 $\frac{du}{dt} = 0,1 \Rightarrow \frac{1}{0,1} du = dt$

$$= 200 \int_0^{12} e^u \cdot 10 du = 2000 e^u \Big|_0^{12} = 2000 \cdot e^{1,2} - 2000 \cdot e^0 + 2000$$

$$= 2000 e^{1,2} - 2000 = 2000 (e^{1,2} - 1)$$

20) O trabalho realizado  $\hat{w}$ :  $W = \int_0^4 (3x^2 - x + 2) dx$ . Calcule.

$$W = \int_0^4 3x^2 dx - \int_0^4 x dx + \int_0^4 2 dx = 3 \cdot \frac{x^3}{3} - \frac{x^2}{2} + 2x \Big|_0^4$$

$$= 4^3 - \frac{4^2}{2} + 2 \cdot 4 = 64 - 8 + 8 = 64 J$$